

SHOPIFY DATASET 2025

Data Cleaning, Exploratory Data Analysis & Time Series Visualization

NAME: VARSHA R B

SUBJECT: PYTHON FOR DATA ANALYSIS (WEEK 4 ASSIGNMENT)

DATASET: SHOPIFY DATASET 2025

COLLAB LINK:

<https://colab.research.google.com/drive/12yIBxLkPVKGo4bBjniHUvtrj7Q9Jpg4O?usp=sharing>

1. Executive Summary

- **Project Scope:** Provides a comprehensive evaluation of the Shopify stock market dataset spanning 2015 to 2025.
- **Key Processes:** Covers end-to-end data processing including cleaning, missing value checks, duplicate detection, date standardization, statistical summary profiling, and time series trend visualization.
- **Environment & Toolchain:** Executed seamlessly in Google Colab utilizing Python core data science modules: Pandas, Seaborn, and Matplotlib.
- **Data Integrity Outcome:** 0 null values and 0 duplicate records identified across a clean sample size of 2,469 total rows.

2. Dataset Overview & Setup

Environment Initialization & Ingestion

- **Library Imports:** Imported foundational data analysis libraries: pandas for data manipulation, seaborn and matplotlib for visualization.
- **Data Source:** Ingested primary dataset 'shopify data 2025.csv' directly into a pandas DataFrame.

```
# Import Core Libraries
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Load Dataset
df = pd.read_csv('shopify data 2025.csv')
```

Dataset Attributes & Schema

- **date (object -> datetime64):** Trading date record formatted with UTC timestamp offsets.
- **open (float64):** Opening stock price recorded at market launch (USD).
- **high (float64):** Maximum recorded price point during trading session (USD).
- **low (float64):** Minimum recorded price point during trading session (USD).
- **close (float64):** Final closing price at market end (USD).

- **adj_close (float64):** Adjusted closing price reflecting stock splits and corporate actions.
- **volume (int64):** Total volume of shares traded throughout the session.

3. Data Cleaning & Preprocessing

Data Hygiene & Audit Results

- **Missing Value Verification:** Executed `df.isnull().sum()` and confirmed 0 missing values across all 7 attributes.
- **Duplicate Audit:** Executed `df.duplicated().sum()` and confirmed 0 duplicate rows within the dataset.
- **Memory Overhead:** Total memory allocation measured at 135.2+ KB, enabling efficient in-memory analytical transformations.

Time-Series Structuring

- **Date Conversion:** Converted string dates into formal pandas Datetime object representations using `pd.to_datetime()`.
- **Chronological Order:** Sorted dataset records in ascending chronological sequence from 2015 to 2025.
- **Index Assignment:** Assigned the standardized date column as the primary index to enable time-series indexing and plotting.

```
# Datetime conversion and indexing
df['date'] = pd.to_datetime(df['date'])
df = df.sort_values('date')
df = df.set_index('date')
```

4. Exploratory Data Analysis (EDA)

Summary Statistics Overview

Metric	Open (\$)	High (\$)	Low (\$)	Close (\$)	Adj Close (\$)	Volume
Count	2,469	2,469	2,469	2,469	2,469	2,469
Mean	48.46	49.51	47.35	48.46	48.46	16,570,491
Std Dev	43.63	44.47	42.64	43.57	43.57	14,369,091
Min	1.94	1.99	1.85	1.93	1.93	1,039,000
25% (Q1)	10.48	10.64	10.28	10.45	10.45	8,098,000
50% (Median)	35.64	36.88	34.66	35.87	35.87	12,758,600
75% (Q3)	76.08	77.17	74.59	75.80	75.80	20,602,000
Max	171.80	176.29	168.51	169.06	169.06	208,959,000

Key Empirical Findings

- **Historical Valuation Expansion:** Closing stock price spans from a historic low of \$1.93 to an all-time peak of \$169.06, reflecting tremendous valuation expansion alongside high price volatility (Standard Deviation = \$43.57).
- **Trading Volume Activity:** Average daily trading activity stands at 16.57 million shares, with maximum single-day trading surges reaching 208.95 million shares during major earnings announcements.
- **Distribution Dynamics:** Median price (\$35.87) remains below the arithmetic mean (\$48.46), confirming right-skewed distribution driven by the historic 2021 market highs.

5. Time Series Visualization

Visualization Script

```
plt.figure(figsize=(12,5))  
  
sns.lineplot(data=df,x=df.index,y="close")  
  
plt.title("Shopify Closing Price Over Time")  
  
plt.xlabel("Date")  
  
plt.ylabel("Closing Price")  
  
plt.show()
```



Historical Trend Analysis Points

- **2015–2019 Steady Accumulation Phase:** Post-IPO period characterized by steady upward growth from \$2.80 to ~\$30.
- **2020–2021 COVID-19 E-Commerce Surge:** Rapid expansion in global digital commerce drove stock valuation to its all-time high above \$169.
- **2022 Post-Pandemic Correction:** Macroeconomic tightening and central bank rate hikes led to a price retraction toward \$30–\$40.

- **2023–2025 Recovery & Growth:** Operational efficiency gains and enterprise adoption restored stock prices above \$90–\$95 by March 2025.

6. Conclusion & Next Steps

Summary of Achievements

- **Pipeline Completion:** Constructed a reliable data pipeline for loading, cleaning, and validating the 2025 Shopify stock dataset.
- **Data Quality:** Confirmed complete data integrity with zero missing values or duplicate records.
- **Statistical Profiling:** Standardized dates and generated comprehensive descriptive statistical profiles.

Future Analysis Roadmap

- **Moving Average Analysis:** Implementation of 50-day and 200-day Simple Moving Average (SMA) indicators to identify technical trend shifts.
- **Predictive Modeling:** Application of ARIMA and Facebook PROPHET models for time-series forecasting and future price prediction.
- **Volatility Profiling:** Evaluation of volatility metrics and trading volume correlation relative to market events.